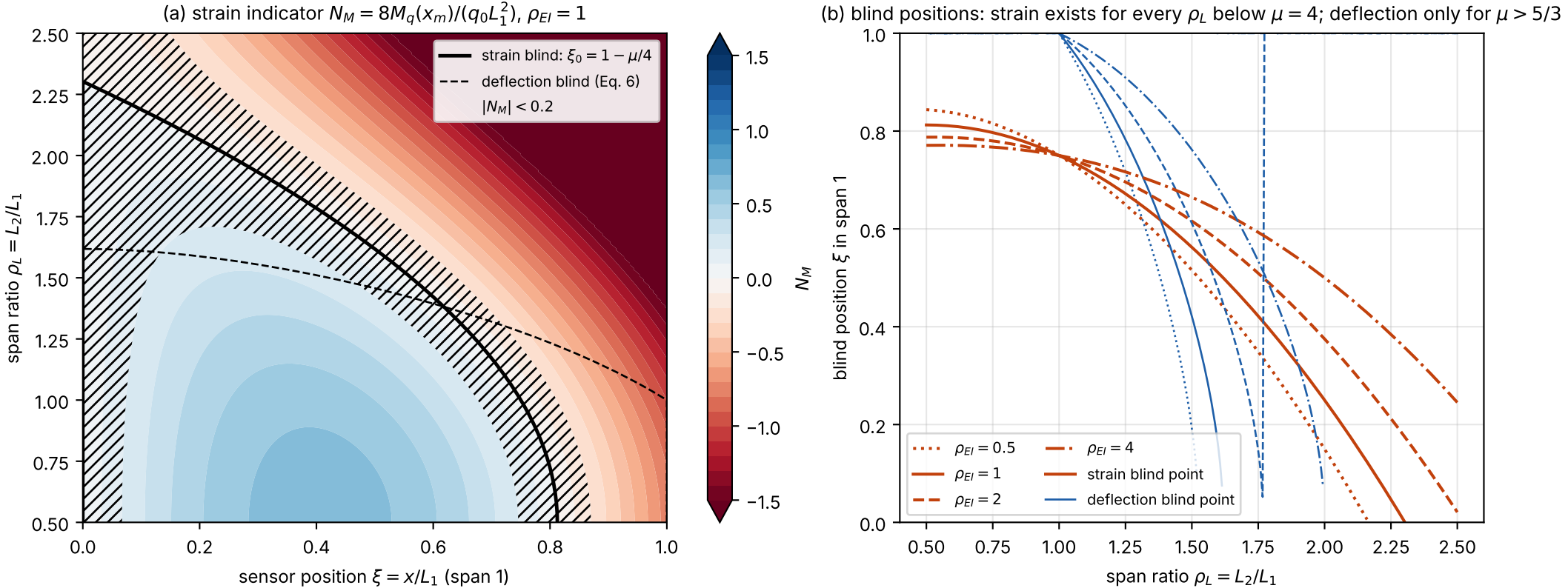

(a) strain indicator $N_M = 8M_q(x_m)/(q_0 L_1^2)$, $\rho_{EI} = 1$
strain blind: $\xi_0 = 1 - \mu/4$
deflection blind (Eq. 6)
$|N_M| < 0.2$
span ratio $\rho_L = L_2/L_1$
sensor position $\xi = x/L_1$ (span 1)
N_M
(b) blind positions: strain exists for every ρ_L below $\mu = 4$; deflection only for $\mu > 5/3$
blind position ξ in span 1
span ratio $\rho_L = L_2/L_1$
$\rho_{EI} = 0.5$
$\rho_{EI} = 1$
$\rho_{EI} = 2$
$\rho_{EI} = 4$
strain blind point
deflection blind point